

Section: **3.1 – STRENGTH OF METALLIC PARTS**
Chapter: **3145**
Title: **BEAM-COLUMNS**
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1 INTRODUCTION

1.1 OBJECT

This chapter is concerned with the strength of uniform beam-columns with a combination of axial and bending loads simultaneously applied.

1.2 METHOD APPLICABILITY

Analysis methodologies described in the present chapter is only applicable for the structural elements (known as *beam-columns*) which have the following characteristics:

- ✓ *Slender and straight* structural elements made of *metallic materials* with *uniform cross-section* (constant cross-section shape along beam-column's length). Typical minimum slenderness ratio is defined in Chapter 3110 'Beams Sections'.
- ✓ Structural elements with axial loads applied in *compression* and *uniform* (locally applied at the beam-column ends only), with transversal loads perpendicularly applied to beam-column's axis (discrete at any point along the beam-column's length and/or distributed along the whole beam-column's length or part of it) and/or with discrete bending moments (applied at any point along the beam-column's length and contained in one of the principal planes of the beam-column's cross-section).

1.3 BEAM-COLUMN FAILURE MODES

A beam-column is a slender structural element with a combination of axial compression and bending loads simultaneously applied. Bending can be caused by the application of eccentric axial loads (not necessarily *centred*, not applied in beam-column's cross-section centroid), transversal loads or due to an initial curvature, or due to a combination of any ones of the mentioned causes.

If axial loads are tension loads, bending moments and deflection caused by transversal loads are reduced. In the other hand, if axial loads are compression loads, maximum bending moment and deflection are increased; therefore, beam-column analysis method must consider the deflection changes caused by applied axial compression loads so, it cannot be considered linear since the effects caused by axial and transversal loads cannot be directly superposed.

The two principal failure modes of beam-columns are the following ones:

- ✓ Buckling failure: If axial compression load is higher than its critical buckling load.
- ✓ Strength failure: If level of internal stresses caused by the combined action of axial compression and bending loads is higher than the beam-column's material allowable value.

Beam-column deflections tend to infinite if applied axial loads cause the buckling of the structural element and beam-column strength failure occurs when deformation at element's fibres exceeds the material ultimate strength; so, both mentioned failure modes must be analysed and, if applied loads do not cause any structural failure, it can be considered that the beam-column is in equilibrium. Only if beam-column is axially loaded in tension, *strength* is the only failure mode that must be analysed (see Chapter 3110 'Beams Sections').

2 METHOD DESCRIPTION

The following paragraphs describe methodologies for calculation of both failure modes of beam-columns, with applied axial compression loads and different types of bending loads.

The following main hypotheses are considered for those analyses:

- ✓ Beam-column is considered to work in *elastic regime*.
- ✓ Bending loads (momentums and/or transverse) are considered to be contained in one of the principal planes of beam-column's cross-section.
- ✓ Strength failure analysis generally considers negligible the effects caused by shear load compared to the effects caused by compression and bending loads since general analysis methodology is only applicable to slender beam-columns (for non-slender beam-columns, shear loads must be calculated by means of deriving the bending moments distribution function and adding the shear loads due caused by torsion loads -if existent- before performing any strength check).
- ✓ Beam-column does not have any intermediate support; supports are considered to be only located at the beam-column ends.

2.1 BEAM-COLUMN BUCKLING

Beam-column buckling failure, with respect to the principal axis of minimum area moment of inertia of its cross-section, is the first failure mode that must be checked. For this analysis, the beam-column must be analysed considering it as a column only loaded with axial compression loads and following the appropriate methodology described in Chapter 3140 'Column Buckling in compression'.

If applied axial compression load is higher than the calculated beam-column's critical buckling load P_F , the beam-column will fail due to buckling. If not, it is considered that the beam-column does not

buckle under its applied loads. Then, the buckling reserve factor is equal to the ratio between beam-column's critical buckling load and applied axial load.

2.2 BEAM-COLUMN STRENGTH

If beam-column is axially loaded in tension or it is axially loaded in compression and, according to §2.1, it has been already checked that no buckling failure occurs, the internal stresses caused by the axial and bending loads must be calculated in order to check the beam-column strength.

If beam-column axial load is in tension, the bending moments and deflections (usually, vertical displacements) caused by transversal loads are reduced so, in this case, the beam-column strength analysis can be conservatively performed by adding the stresses caused by transversal loads to the stresses caused by tensile loads, which must be previously and separately calculated by classical analysis methods (more information included in Chapter 3110 'Beams Sections').

If beam-column axial load is in compression, the bending moments and deflections caused by transversal loads are increased so, direct superposition of the effects caused by each type of loads is not conservative and the influence of compression on the transversal loads must be taken into account.

The present section describes the methodology for calculation of the maximum bending moment at beam-column caused by the combined action of compression and bending loads. Once this bending moment is obtained, values of maximum tension and compression stress at beam-column can be calculated with the following expression and compared with the corresponding material's strength allowable values:

$$\sigma_{\max} = \frac{P}{A} \pm \frac{M_{\max}}{I} \cdot c$$

where:

- $\sigma_{\max}$ is the maximum tension or compression axial stress in the beam-column [MPa].
- $M_{\max}$ is the calculated maximum bending moment in the beam-column [N·mm].
- P is the axial compression load applied in the beam-column ends [N].
- A is the area of beam-column's cross-section [mm²].
- I is the moment of inertia of beam-column's cross-section respect to the principal plane where bending effects are being analysed [mm⁴].
- c is the distance between beam-column's neutral axis and its cross-section outer fibre [mm].

Then, beam-column's strength reserve factor is equal to the ratio between allowable stress (whatever corresponds: F_{tu} , crippling stress, etc.) and calculated maximum axial stress $\sigma_{\max}$.

According to Ref. 2, the maximum bending moment $M_{\max}$ of uniform beam-columns with compression axial loads can be calculated by means of the expressions included in Table 1 to Table 5 for the most frequent load cases. Expressions for calculation of maximum bending point location at the beam-column ($x_{M_{\max}}$) are also provided.

Notice that parameter k of Table 1 to Table 10 is calculated as (see Ref. 2):

$$k = \sqrt{\frac{P}{E_c \cdot I}}$$

where:

- P is the axial compression load applied in the beam-column ends [N].
- E_c is the compression young modulus of beam-column's material [MPa].
- I is the area moment of inertia of beam-column's cross-section respect to the principal plane where bending effects are being analysed [mm⁴].

If a beam-column loaded with a compression axial load P has several transversal loads applied, resulting bending moments and deflections can be calculated by superposing the ones separately calculated for each load sub-case in combination with the compression axial load P .

Notice that if beam-column loading conditions can be separated in several less complex sub-cases, which do not obtain the maximum bending moment at the same position of beam-column, the maximum bending moment for the combined load case cannot be directly calculated by summing up the values of all the maximum moments calculated for each simple sub-case since the final result would be too conservative. Additionally, if the values of the maximum bending moments calculated for the simple sub-cases have different signs, the maximum bending moment of combined load case will not be equal to the difference between the maximum moments obtained for the sub-cases. Therefore, in those mentioned situations, it is necessary to obtain the bending moments distribution for the combined load case in order to obtain the maximum bending moment at the analysed beam-column.

Then, the general bending moments distribution of a beam-column can be defined as (see Ref. 1 and Ref. 2):

$$M(x) = \sum_{i=0}^n A_i \sin(kx) + \sum_{i=0}^n B_i \cos(kx) + \frac{1}{k^2} \cdot \sum_{i=0}^n f_i(x)$$

where:

- M is the bending moment of beam-column [N·mm].
- x is the longitudinal coordinate of beam-column's reference system [mm].
- n is the total quantity of simple sub-cases i in which can be separated the whole analysed load case.
- A_i, B_i are constant coefficients associated to each sub-case i [N·mm].
- $f_i(x)$ is a function associated to each sub-case i [N/mm].
- k is an analysis parameter already defined in previous paragraphs [mm⁻¹].

The coefficients A_i and B_i and function $f_i(x)$ associated to each sub-case i can be calculated using the expressions contained from Table 6 to Table 10, which were defined in Ref. 2 from Ref. 1.

Once the $M(x)$ distribution is defined for the analysed beam-column, the internal shear loads V , cross-section gyration angle θ and vertical displacement/deflection δ can be also calculated at any location x of the beam-column according to the following expressions:

$$V = \frac{dM}{dx} \quad \theta = -\frac{1}{E_c I} \int M dx + C_1 \quad \delta = \int \theta dx + C_2$$

where:

- E_c is the material's compression elasticity modulus [MPa].
- I is the area moment of inertia of beam-column's cross-section respect to the principal plane where bending effects are being analysed [mm⁴].
- C_1, C_2 are integration constants to be determined according to the beam-column boundary conditions [N·mm].

Notice that sign criterion considered in all above-mentioned expressions and in the equations shown from Table 1 to Table 10 is the following one:

- ✓ Origin of coordinate x is located at the beam-column's left end and x positive sign is following to the right's end direction.
- ✓ All angles are expressed in radians.
- ✓ Positive sign of applied loads and moments correspond to the signs shown in figures.
- ✓ Bending moments are positive if they cause compression in the fibres located above the neutral axis of beam-column's cross section.
- ✓ Beam-column's deflections δ (usually, vertical displacements) are considered positive if they go downwards.

2.2.1 Alternative analysis methods:

The method described in present chapter is an explicit solution but could exist situations where complexity of the analytical equations justify performing the analysis by means of other methods. According to performed checks, the Newmark's numerical method, often used in non-uniform beam-columns analyses, has demonstrated to obtain very similar results (bending moments deviation < 2%) so, it can be considered a numerical alternative method for the explicit one described in present chapter.

For more information of Newmark method, see Chapter 3141 'Buckling of non-uniform columns'.

BEAM-COLUMNS

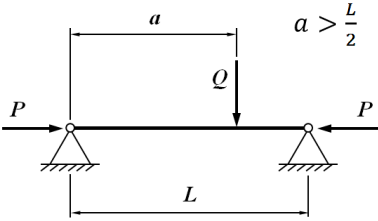
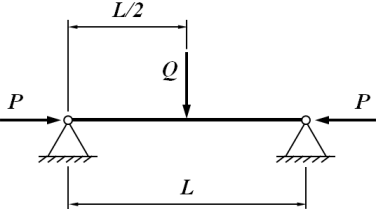
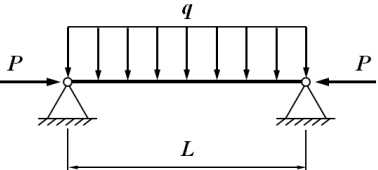
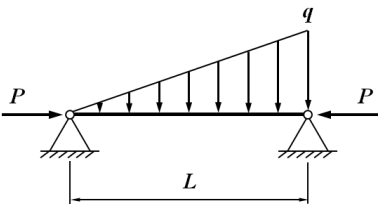
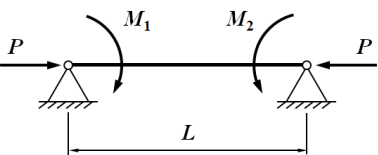
Sub-case		$M_{\max}$		$x_{M_{\max}}$
#1		$a < \frac{\pi}{2k}$	$\frac{Q \sin[k(L-a)]}{k \sin(kL)} \sin(ka)$	a
		$a \geq \frac{\pi}{2k}$	$\frac{Q \sin[k(L-a)]}{k \sin(kL)}$	$\frac{\pi}{2k}$
#2		$\frac{Q}{2k} \tan \frac{kL}{2}$		$\frac{L}{2}$
#3		$\frac{q}{k^2} \left(\frac{1}{\cos \frac{kL}{2}} - 1 \right)$		$\frac{L}{2}$
#4		$\frac{q}{k^2} \left(\frac{\sin(kx)}{\sin(kL)} - \frac{x}{L} \right)$		$\frac{1}{k} \cos^{-1} \frac{\sin(kL)}{kL}$
#5		$\frac{\sqrt{M_1^2 - 2M_1M_2 \cos(kL) + M_2^2}}{\sin(kL)}$		$\frac{1}{k} \tan^{-1} \frac{A}{B}$ $A = M_2 - M_1 \cos(kL)$ $B = M_1 \sin(kL)$

Table 1. Beam-column's maximum bending moment for different loading & supporting conditions.

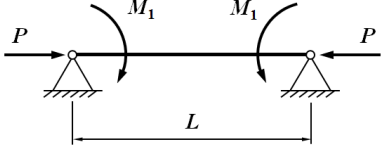
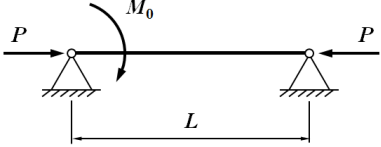
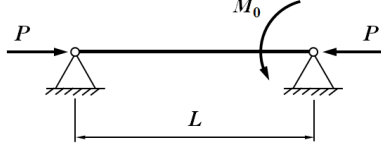
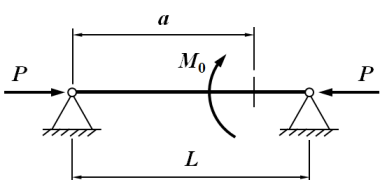
Sub-case		$M_{\max}$		$x_{M_{\max}}$
#6		$\frac{M_1}{\cos \frac{kL}{2}}$		$\frac{L}{2}$
#7		$L < \frac{\pi}{2k}$	M_0	0
		$L \geq \frac{\pi}{2k}$	$\frac{M_0}{\sin(kL)}$	$L - \frac{\pi}{2k}$
#8		$L < \frac{\pi}{2k}$	M_0	L
		$L \geq \frac{\pi}{2k}$	$\frac{M_0}{\sin(kL)}$	$\frac{\pi}{2k}$
#9		$a < \frac{\pi}{2k}$	$-\frac{M_0 \sin[k(L-a)] \cos(ka)}{\sin(kL)} + \frac{M_0 \cos[k(L-a)] \sin(ka)}{\sin(kL)}$	a
		$a \geq \frac{\pi}{2k}$	$\frac{M_0 \cos[k(L-a)]}{\sin(kL)}$	$\frac{\pi}{2k}$

Table 2. Beam-column's maximum bending moment for different loading & supporting conditions.

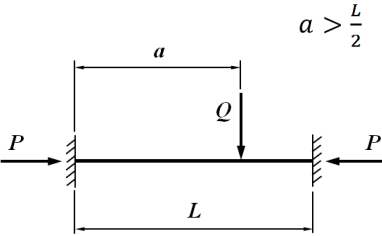
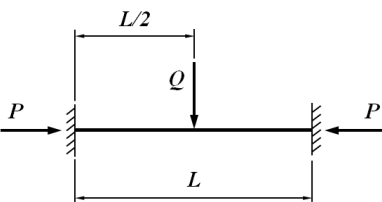
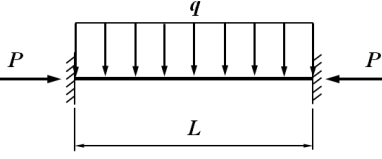
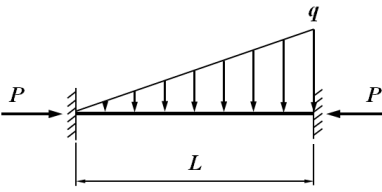
Sub-case	$M_{\max}$	$x_{M_{\max}}$
#10  $a > \frac{L}{2}$	$M^{(-)} = -\frac{Q}{k} \frac{C_4 C_{a3} - C_3 C_{a4}}{C_3^2 - C_2 C_4}$ $C_2 = \sin(kL)$ $C_3 = 1 - \cos(kL)$ $C_4 = kL - \sin(kL)$ $C_{a3} = 1 - \cos(ka)$ $C_{a4} = ka - \sin(ka)$	L
#11 	$M^{(-)} = -\frac{Q}{2k} \tan \frac{kL}{4}$	$0, L$
	$M^{(+)} = +\frac{Q}{2k} \tan \frac{kL}{4}$	$\frac{L}{2}$
#12 	$M^{(-)} = -\frac{q}{k^2} \left(1 - \frac{\frac{kL}{2}}{\tan \frac{kL}{2}} \right)$	$0, L$
	$M^{(+)} = +\frac{q}{k^2} \left(\frac{\frac{kL}{2}}{\sin \frac{kL}{2}} - 1 \right)$	$\frac{L}{2}$
#13 	$M^{(-)} = -\frac{q}{k^2} \left(1 - \frac{\frac{A}{2} \sin A - \frac{A^2}{6} - \frac{A^2}{3} \cos A}{2 - 2 \cos A - A \sin A} \right)$ $A = kL$	L

Table 3. Beam-column's maximum bending moment for different loading & supporting conditions.

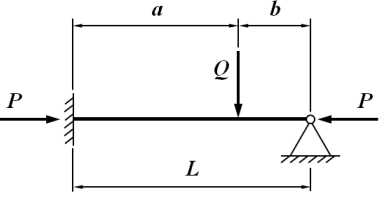
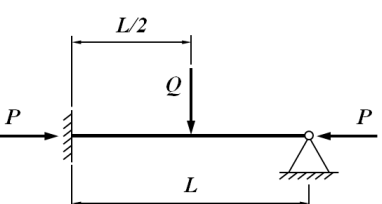
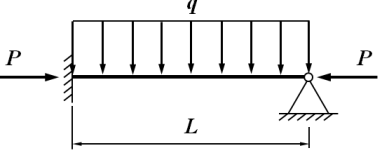
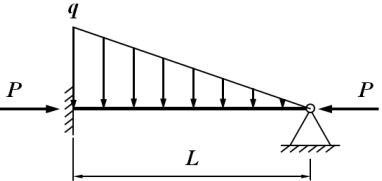
Sub-case		$M_{\max}$	$x_{M_{\max}}$
#14		$M^{(-)} = -\frac{Q}{k} \frac{kL \sin(kb) - kb \sin(kL)}{\sin(kL) - kL \cos kL}$	0
		$M^{(+)} = +\frac{Q}{k} \frac{kb + \sin(ka) - kL \cos(ka)}{\sin(kL) - kL \cos kL} \sin(kb)$	a
#15		$M^{(-)} = -QL \frac{1 - \cos \frac{kL}{2}}{\sin(kL) - kL \cos(kL)} \sin \frac{kL}{2}$	0
		$M^{(+)} = \frac{Q}{k} \frac{\frac{kL}{2} + \sin \frac{kL}{2} - kL \cos \frac{kL}{2}}{\sin(kL) - kL \cos(kL)} \sin \frac{kL}{2}$	$\frac{L}{2}$
#16		$M^{(-)} = -\frac{qL}{k} \frac{\tan \frac{kL}{2} - \frac{kL}{2}}{\tan(kL) - kL} \tan(kL)$	0
#17		$M^{(-)} = -\frac{q}{k^2} \frac{\left(1 - \frac{k^2 L^2}{3}\right) \tan(kL) - kL}{\tan(kL) - kL}$	0

Table 4. Beam-column's maximum bending moment for different loading & supporting conditions.

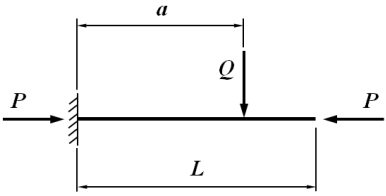
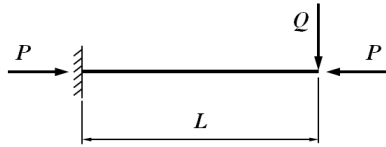
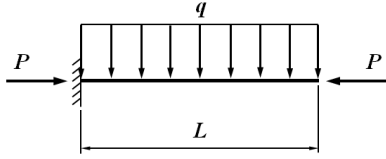
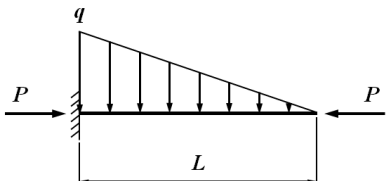
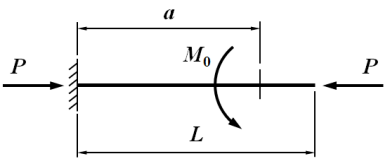
Sub-case		$M_{\max}$	$x_{M_{\max}}$
#18		$-\frac{Q}{k} \{ \sin(ka) + [1 - \cos(ka)] \tan(kL) \}$	0
#19		$-\frac{Q}{k} \tan(kL)$	0
#20		$-\frac{q}{k^2} \left[1 + kL \tan(kL) - \frac{1}{\cos(kL)} \right]$	0
#21		$-\frac{q}{k^2} \left[1 + \frac{kL}{2} \tan(kL) - \frac{\tan(kL)}{kL} \right]$	0
#22		$\frac{\cos[k(L-a)]}{\cos(kL)} M_0$	0

Table 5. Beam-column's maximum bending moment for different loading & supporting conditions.

Sub-case	A		B	f(x)
#1	$x < a$	$\frac{Q}{k} \frac{\sin[k(L-a)]}{\sin(kL)}$	0	0
	$x > a$	$-\frac{Q}{k} \frac{\sin(ka)}{\tan(kL)}$	$\frac{Q}{k} \sin(ka)$	0
#2	$x < \frac{L}{2}$	$\frac{Q}{2k} \frac{1}{\cos \frac{kL}{2}}$	0	0
#3	$\frac{q}{k^2} \tan \frac{kL}{2}$		$\frac{q}{k^2}$	$-q$
#4	$\frac{q}{k^2} \frac{1}{\sin(kL)}$		0	$-\frac{q}{L} x$
#5	$\frac{M_2 - M_1 \cos(kL)}{\sin(kL)}$		M_1	0

Table 6. Bending moments law coefficients for different loading & supporting conditions.

Sub-case	A		B	f(x)
#6	$M_1 \tan \frac{kL}{2}$		M_1	0
#7	$-\frac{M_0}{\tan(kL)}$		M_0	0
#8	$\frac{M_0}{\sin(kL)}$		0	0
#9	$x < a$	$-\frac{M_0 \cos[k(L-a)]}{\sin(kL)}$	0	0
	$x > a$	$-\frac{M_0 \cos(ka)}{\tan(kL)}$	$M_0 \cos(ka)$	0

Table 7. Bending moments law coefficients for different loading & supporting conditions.

Sub-case	A		B	f(x)
#10	$x < a$	$\frac{Q}{k} \frac{C_3 C_{a3}^* - C_2 C_{a4}^*}{C_3^2 - C_2 C_4}$	$-\frac{Q}{k} \frac{C_4 C_{a3}^* - C_3 C_{a4}^*}{C_3^2 - C_2 C_4}$	0
	$x > a$	$A_{(x < a)} - \frac{Q}{k} \cos(ka)$	$B_{(x < a)} + \frac{Q}{k} \sin(ka)$	0
	$C_2 = \sin(kL)$ $C_3 = 1 - \cos(kL)$ $C_4 = kL - \sin(kL)$ $C_{a3}^* = 1 - \cos[k(L - a)]$ $C_{a4}^* = k(L - a) - \sin[k(L - a)]$			
#11	$x < \frac{L}{2}$	$\frac{Q}{2k}$	$-\frac{Q}{2k} \tan \frac{kL}{4}$	0
#12	$\frac{qL}{2k}$		$\frac{qL}{2k} \frac{1}{\tan \frac{kL}{2}}$	$-q$
#13	$\frac{qL}{6k} \frac{3 - 3 \cos(kL) - kL \sin(kL)}{2 - 2 \cos(kL) - kL \sin(kL)}$		$-\frac{qL}{6k} \frac{2kL + kL \cos(kL) - 3 \sin(kL)}{2 - 2 \cos(kL) - kL \sin(kL)}$	$-\frac{q}{L} x$

Table 8. Bending moments law coefficients for different loading & supporting conditions.

Sub-case	A		B	f(x)
#14	$x < a$	$-\frac{Q}{k} \frac{kb \cos(kL) - \sin(kb)}{\sin(kL) - kL \cos(kL)}$	$\frac{Q}{k} \frac{kb \sin(kL) - kL \sin(kb)}{\sin(kL) - kL \cos(kL)}$	0
	$x > a$	$-\frac{Q}{k} \frac{kb + \sin(ka) - kL \cos(ka)}{\sin(kL) - kL \cos(kL)} \cos(kL)$	$\frac{Q}{k} \frac{kb + \sin(ka) - kL \cos(ka)}{\sin(kL) - kL \cos(kL)} \sin(kL)$	0
#15	$x < \frac{L}{2}$	$-\frac{Q}{k} \frac{\frac{kL}{2} \cos(kL) - \sin \frac{kL}{2}}{\sin(kL) - kL \cos(kL)}$	$\frac{Q}{k} \frac{\frac{kL}{2} \sin(kL) - kL \sin \frac{kL}{2}}{\sin(kL) - kL \cos(kL)}$	0
#16	$\frac{qL}{k} \frac{\tan \frac{kL}{2} - \frac{kL}{2}}{\tan(kL) - kL} + \frac{q}{k^2} \tan \frac{kL}{2}$		$-\frac{qL}{k} \frac{\tan \frac{kL}{2} - \frac{kL}{2}}{\tan(kL) - kL} \tan(kL) + \frac{q}{k^2}$	$-q$
#17	$-\frac{qL}{6k} \frac{2kL}{\tan(kL) - kL}$		$\frac{qL}{6k} \frac{2kL \tan(kL)}{\tan(kL) - kL}$	$-\frac{q}{L}(L - x)$

Table 9. Bending moments law coefficients for different loading & supporting conditions.

Sub-case	A		B	f(x)
#18	$x < a$	$\frac{Q}{k}$	$-\frac{Q}{k} \{ \sin(ka) + [1 - \cos(ka)] \tan(kL) \}$	0
	$x > a$	$\frac{Q}{k} [1 - \cos(ka)]$	$-\frac{Q}{k} [1 - \cos(ka)] \tan(kL)$	0
#19	$x < L$	$\frac{Q}{k}$	$-\frac{Q}{k} \tan(kL)$	0
#20	$\frac{qL}{k}$		$\frac{q}{k^2} \left[\frac{1}{\cos(kL)} - kL \tan(kL) \right]$	$-q$
#21	$\frac{q}{k^2} \left(\frac{kL}{2} - \frac{1}{kL} \right)$		$-\frac{q}{k^2} \left(\frac{kL}{2} - \frac{1}{kL} \right) \tan(kL)$	$-\frac{q}{L} (L - x)$
#22	$x < a$	0	$\frac{\cos[k(L - a)]}{\cos(kL)} M_0$	0
	$x > a$	$-M_0 \sin(ka)$	$M_0 \sin(ka) \tan(kL)$	0

Table 10. Bending moments law coefficients for different loading & supporting conditions.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A	mm ²	Area of beam-column's cross-section
δ	mm	Beam-column's deflection (vertical displacement)
E	MPa	Material's elasticity modulus
E_c	MPa	Material's compression elasticity modulus
F_{tu}	MPa	Material's ultimate tensile stress
I	mm ⁴	Minimum moment of inertia of beam-column's cross-section
L	mm	Beam-column's length
M	MPa	Bending moment
P	N	Axial load
P_F	N	Critical buckling failure load
σ	MPa	Generic failure axial stress
θ	rad	Beam-column's cross-section gyration angle
V	N	Shear load
x	mm	Beam- column's longitudinal coordinate

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[S. P. Timoshenko & J. M. Gere]	2 nd Ed.	"Theory of Elastic Stability"
Ref. 2	[C.A.S.A.] Manual de Diseño	23/May/2000	"Manual de Diseño. Volumen 08. Cálculo Estructural. Sección XIV. Capítulo 144" CASA technical handbook (original in Spanish).

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
PANCO (Airbus DS)	4.2	This program analyses the column-buckling of variable cross-section beams and performs the beam-column strength checks for different column's supporting conditions.

Table 11. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernaß Daoud	X	
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
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Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 12. Distribution list (of electronic document, ahead of public release)

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 02/12/2019	Guillermo Moreda	First issue	All pages

Table 13. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods (TEASP-TL3)	G. Baños 02/12/2019	F. Glock 02/12/2019

Table 14. Approval signatures table for last revision of this chapter